Scalar Signorini in dual-mixed form on a Bernstein cone: MRV postprocessing and a contact-boundary estimator

Working note

November 13, 2026 (version 2)

Abstract

The scalar Signorini problem is discretised in dual-mixed form with $\mathrm{BDM}_{k+1} \times \mathrm{P}_k$, $k = 0, \dots, 3$, without a Lagrange multiplier: the sign of the normal flux is imposed on the Bernstein coefficients of its trace on each contact edge, and the resulting variational inequality is solved by a primal-dual active set acting coefficient by coefficient. The solution is postprocessed with the local residual minimisation of Muga–Rojas–Vega (MRV) [4], and an a posteriori indicator is built from the improved MRV estimator plus a contact-boundary term weighted by the Bernstein partition of unity of the discrete active set. The indicator does not use the equilibrium equation. Its reliability is not proved; everything said about it here is numerical. The experiments cover manufactured solutions with smooth and singular transition points, the problem without contact as a control, and the examples of Führer–Heuer–Stephan [2] and Bartels–Gudi–Kaltenbach [1].

Changes with respect to version 1.

- The two open questions on the indicator (Section 8) are stated as hypotheses with the evidence for and against them. Version 1 stated as facts that (i) an indicator that does not use the equilibrium equation cannot see the oscillation of f, and (ii) with the transition point inside an edge the missing error is contact error that no term captures and that adaptivity removes. A new control experiment, the same solution without contact (E5), shows that part of (ii) is not a contact effect.
- Corrected details: the edge-wise classification also improves on having no contact term; the complementarity term overestimates by up to ~ 50 (not ~ 30); the contact term follows the contact-zone trace error within a factor 0.97–2.2; Theorem 3.9 of [4] needs the saturation assumption also for the linear problem.
- Every claim on the indicator is backed by a table or a figure.

1 Problem and notation

Let $\Omega \subset \mathbb{R}^2$ be a polygon with $\partial\Omega = \Gamma_D \cup \Gamma_C$, $\Gamma_D \cap \Gamma_C = \emptyset$, $|\Gamma_D| > 0$, and let $f \in L^2(\Omega)$, $u_D \in H^{1/2}(\Gamma_D)$. The problem is

$$-\Delta u = f \text{ in } \Omega, \qquad u = u_D \text{ on } \Gamma_D, \qquad u \geq 0, \quad \partial_n u \geq 0, \quad u\,\partial_n u = 0 \text{ on } \Gamma_C. \tag{1}$$

We write $\sigma := \nabla u$, so that $\operatorname{div}\sigma = -f$ and the sign condition reads $\sigma \cdot n \geq 0$ on Γ_C (dual-mixed form as in [6]); in the notation of [4], $q := -\sigma$. The contact set is $\Gamma_C^0 := \{u = 0\} \subset \Gamma_C$ and a *transition point* is an endpoint of Γ_C^0 in the relative interior of Γ_C.

$\mathcal{T}_h$ is a shape-regular triangulation compatible with Γ_D/Γ_C; $\mathcal{F}_h^i, \mathcal{F}_h^D, \mathcal{F}_h^C$ are the interior edges and the edges on Γ_D and Γ_C, and K_E is the element containing $E \in \mathcal{F}_h^C$. For $E \in \mathcal{F}_h^C$ and $g \in \mathrm{P}_m(E)$, $b_{E,i}(g)$, $i = 0, \dots, m$, are the degree-m Bernstein coefficients of g and $B_{E,i}$ the Bernstein basis, so that $g = \sum_i b_{E,i}(g)\, B_{E,i}$, $B_{E,i} \geq 0$, $\sum_i B_{E,i} = 1$.

Remark 1.1 (obstacle, Neumann, reaction)**.** With an obstacle $u \geq \chi$ on Γ_C, a homogeneous Neumann part Γ_N ($\sigma \cdot n = 0$) and a reaction term $-\Delta u + cu = f$, $c \geq 0$, everything below holds after replacing u and ν_h by $u - \chi$ and $\nu_h - \chi$ on Γ_C, imposing $\sigma_h \cdot n = 0$ on Γ_N, and adding $-(c\, u_h, v)$ to the equilibrium equation. If $c > 0$, Γ_D may be empty (the examples of [2] have $c = 1$ and Signorini conditions on all of $\partial\Omega$).

2 Mixed solution

Let $\Sigma_h = \mathrm{BDM}_{k+1}$ and $Q_h = \mathrm{P}_k(\mathcal{T}_h)$, $k \geq 0$ (MRV's p is $k+1$). The normal trace of $\tau \in \Sigma_h$ on $E \in \mathcal{F}_h^C$ lies in $\mathrm{P}_{k+1}(E)$ and is fixed by the $k+2$ degrees of freedom of E. The discrete cone is

$$K_h := \{\tau \in \Sigma_h :\ b_{E,i}(\tau \cdot n|_E) \geq 0 \quad \forall E \in \mathcal{F}_h^C,\ i = 0, \dots, k+1\},$$

and $(\sigma_h, u_h) \in K_h \times Q_h$ solves

$$(\sigma_h, \tau - \sigma_h) + (u_h, \mathrm{div}(\tau - \sigma_h)) \geq \langle u_D, (\tau - \sigma_h)\cdot n\rangle_{\Gamma_D} \quad \forall \tau \in K_h, \qquad (\mathrm{div}\,\sigma_h, v) = -(f, v) \quad \forall v \in Q_h. \tag{2}$$

Since the Bernstein basis is non-negative, $\tau \in K_h$ implies $\tau \cdot n \geq 0$ pointwise on Γ_C. For $k = 0$ the condition is also necessary (the trace is linear and its coefficients are its end values); for $k \geq 1$ it is only sufficient. Hence σ_h satisfies the sign condition at every point, and $\mathrm{div}\,\sigma_h = -Q_h f$ with Q_h the L^2 projection onto $\mathrm{P}_k(\mathcal{T}_h)$.

Remark 2.1 (cone without multiplier)**.** On each $E \in \mathcal{F}_h^C$ the $k+2$ degrees of freedom of $\sigma_h \cdot n|_E$ are replaced by its Bernstein coefficients $c_{E,i}$ through a fixed $(k+2) \times (k+2)$ matrix, computed from the values of the basis at the Bernstein nodes. Each coefficient is either *in contact* (free) or *separated* ($c_{E,i} = 0$). This is a linear change of variables $x = T\tilde{x}$ applied to the assembled saddle-point system, and no Lagrange multiplier is introduced.

Remark 2.2 (active set)**.** The multiplier of $c_{E,i} \geq 0$ is $\mu_{E,i} := \left[T^\top (Kx - b)\right]_{E,i}$, the residual of the full system in the direction of the flux whose normal trace on E is $B_{E,i}$; its continuous counterpart is $\int_E u\, B_{E,i}$. The KKT conditions of (2) are $c \geq 0$, $\mu \geq 0$, $c_{E,i}\mu_{E,i} = 0$, and the primal-dual active set update is

$$\text{contact} \to \text{separated if } c_{E,i} < -\mathrm{tol}\,\|c\|_\infty, \qquad \text{separated} \to \text{contact if } \mu_{E,i} < -\mathrm{tol}\,\|\mu\|_\infty,$$

with $\mathrm{tol} = 10^{-10}$ (cf. [3]). It terminated in all runs, in at most 44 iterations ($k = 3$, $N \approx 5 \cdot 10^4$), starting from the fully separated state. We write $\beta_E := \sum_{i \text{ in contact}} B_{E,i}$, the Bernstein partition of unity restricted to the coefficients in contact; $\beta_E \equiv 1$ on an edge fully in contact, $\beta_E \equiv 0$ on a separated one, and in between on an edge that a transition point cuts.

3 Postprocessing

With $q_h := -\sigma_h$, on each $K \in \mathcal{T}_h$ the MRV problem [4, (3.2)] reads: find $(\varepsilon_K, \nu_K) \in V_{\star,K}^{k+3} \times V_K^{k+2}$ with

$$\begin{aligned} (\nabla\varepsilon_K + \nabla\nu_K, \nabla v)_K &= -(q_h, \nabla v)_K \quad \forall v \in V_{\star,K}^{k+3}, \\ (\nabla w, \nabla\varepsilon_K)_K &= 0 \qquad\qquad\quad \forall w \in V_{\star,K}^{k+2}, \qquad (\nu_K, 1)_K = (u_h, 1)_K, \end{aligned} \tag{3}$$

where $V_K^r = \mathrm{P}_r(K)$ and $V_{\star,K}^r$ its zero-mean subspace. We write $\nu_h|_K := \nu_K$, $\varepsilon_h|_K := \varepsilon_K$ and $\tilde{\eta}_K := \|\nabla\varepsilon_K\|_K$. The postprocessing is not constrained on Γ_C: $\nu_h \geq 0$ may fail there.

Remark 3.1 (Stenberg)**.** By [4, Prop. 3.1], ν_h coincides with the (modified) Stenberg postprocessing [4, (2.10)]. Both were computed independently on the meshes of E1: the relative difference is at most $5 \cdot 10^{-15}$ in L^2 and 10^{-13} in the broken H^1 seminorm, for $k = 0, \dots, 3$.

4 Indicator

Definition 4.1. For $K \in \mathcal{T}_h$,

$$\eta_K^2 := \tilde{\eta}_K^2 + \|\sigma_h - \nabla\nu_K\|_K^2 + \tfrac{1}{2}\sum_{F\in\mathcal{F}_K^i} h_F^{-1}\|[\![\nu_h]\!]\|_F^2 + \sum_{F\in\mathcal{F}_K^D} h_F^{-1}\|u_D - \nu_h\|_F^2$$
$$+ \sum_{E\in\mathcal{F}_K^C}\Big(h_E^{-1}\|\beta_E^{1/2}\,\nu_h\|_E^2 + h_E^{-1}\|(1-\beta_E)^{1/2}\min(\nu_h,0)\|_E^2\Big),$$

and $\eta^2 := \sum_K \eta_K^2$. With an obstacle, ν_h is replaced by $\nu_h - \chi$ in the last two terms (Remark 1.1).

On Γ_C the trace of u is unknown, and the Dirichlet term of [4, (3.9)] is replaced by a trace term where the discrete active set says contact ($u = 0$) and by the violation of $u \geq 0$ where it says separation. Table 1 compares the terms with those of MRV.

term	MRV, estimator (3.9)	this note
$\tilde{\eta}_K = \|\nabla\varepsilon_K\|_K$	yes	yes
$\|q_h + \nabla\nu_K\|_K = \|\sigma_h - \nabla\nu_K\|_K$	yes	yes
jumps $h_F^{-1}\|[\![\nu_h]\!]\|_F^2$	yes	yes
Dirichlet $h_F^{-1}\|u_D - \nu_h\|_F^2$	yes	yes
contact boundary	absent	β_E-weighted trace and $\min(\nu_h, 0)$
sign of $\sigma_h \cdot n$	absent	not needed: $\sigma_h \cdot n \geq 0$ pointwise
equilibrium residual / oscillation	osc(q), via Fortin	not used

Table 1: Comparison with the improved estimator (3.9) of [4].

Remark 4.2 (what is not included)**.** (a) No equilibrium residual: the indicator uses only the constitutive law $q = -\nabla u$, the conformity of ν_h and the boundary conditions, as MRV's. (b) No complementarity term $|\langle\sigma_h \cdot n, \nu_h\rangle_E|$: it is the product of an $O(1)$ factor and an O(error) factor, not quadratic in the error, and it overestimates more and more as k grows (Section 6.3). (c) No term on Γ_N, where $\sigma_h \cdot n = 0$ is exact.

Remark 4.3 (reliability)**.** Reliability for the variational inequality is not proved. Even for the linear problem, Theorem 3.9 of [4] gives it under the saturation assumption (Assumption 3.5), with constant $1/(1-\delta)$, where $\delta := \|\nabla_h(u-\theta_h)\|/\|\nabla_h(u-\nu_h)\|$ and θ_h is the Stenberg postprocessing of degree $k+3$ [4, (3.3)].

5 Error measures

The error of the postprocessing is measured in the MRV norm with jumps, with a trace term only where the trace of u is prescribed:

$$E_s^2 := \|\sigma-\sigma_h\|^2 + \|\nabla_h(u-\nu_h)\|^2 + \sum_{F\in\mathcal{F}_h^i} h_F^{-1}\|[\![\nu_h]\!]\|_F^2 + \sum_{F\in\mathcal{F}_h^D} h_F^{-1}\|u-\nu_h\|_F^2 + \sum_{E\in\mathcal{F}_h^C} h_E^{-1}\|u-\nu_h\|_{E\cap\Gamma_C^0}^2.$$

On the separated part of Γ_C, as on Γ_N, the trace of u is not prescribed and the norm carries no trace term. The built-in indicator $\tilde{\eta} = (\sum_K \tilde{\eta}_K^2)^{1/2}$ is measured against its own norm [4, Thm. 3.6], $E^2 := \|\nabla_h(u-\nu_h)\|^2 + \|q - q_h\|_*^2$, where $\|\cdot\|_*$ is the discrete dual norm over $\nabla V_{\star,K}^{k+3}$, computed exactly through its local Riesz representative. Effectivities are η/E_s and $\tilde{\eta}/E$.

6 Numerical experiments

Implementation. Julia prototype on Gridap, with $\mathrm{BDM}_{k+1} \times \mathrm{P}_k$, $k = 0, \dots, 3$, on meshes of right triangles (each square of a Cartesian grid split along one diagonal). Uniform refinement halves h; adaptive refinement uses Dörfler marking ($\theta = 1/2$) on η_K^2 and newest-vertex bisection. The local problems (3) are solved densely element by element. N is the number of degrees of freedom of the mixed problem. Rates are least-squares slopes over the last decade of N, reported as $e \sim N^s$ (in 2D, $h \sim N^{-1/2}$); effectivity ranges are over the same decade.

Examples. $u_s := -r^\alpha \sin(\alpha\theta)$ in polar coordinates centred at $(x_0, 0)$, $\theta \in [0, \pi]$; on $y = 0$ it has contact for $x > x_0$ and separation for $x < x_0$.

- E1 $u = u_s + y^2$, $\alpha = 11/2$, $x_0 = 0$, on $(-\frac{1}{2}, \frac{1}{2}) \times (0, 1)$, Γ_C the bottom, exact Dirichlet data elsewhere, $f = -2$. Smooth: $u \in H^{13/2-\epsilon}$.
- E2 As E1 with $\alpha = 3/2$: $u \in H^{5/2-\epsilon}$, the generic Signorini regularity at a transition point. $x_0 = 0$ is a node of every mesh.
- E3 As E2 with $x_0 = 1/(2\pi)$: irrational, so never a node of these dyadic meshes; the transition always lies inside an edge.
- E4 E3 plus $L = 0.3x - 0.05$, with obstacle $\chi = L$ on Γ_C (L is harmonic and $\partial_y L = 0$, so $u - \chi$ and $\sigma \cdot n$ on Γ_C are those of E3).
- E5 Control: the solutions of E2 and E3 *without contact* (Dirichlet data on the whole boundary), with the same indicator (it reduces to MRV's (3.9)).
- E6 FHS 4.2 [2]: $-\Delta u + u = f$ on $(-1, 1) \times (0, 1)$, Signorini on all of $\partial\Omega$, piecewise polynomial u; mesh aligned with $x = 0$, transitions at nodes.
- E7 FHS 4.3 [2]: L-shaped domain, $-\Delta u + u = f$, $f = -1$ for $|x| \le 0.8$ and $\frac{1}{2}$ else, Signorini on all of $\partial\Omega$. No exact solution.
- E8 BGK 7.2 [1]: $(0, 1)^2$, $u = -10\,\psi(r)\, r^{3/2} \sin(\frac{3}{2}\theta)$ centred at $(\frac{1}{2}, 0)$, ψ the C^4 cut-off of degree 9 on $[0, 0.45]$; $f \notin \mathrm{P}_k$ with large gradients. The transition is a node.
- E9 BGK 7.3.1 [1]: $(-1, 1)^2$, $f = -1$, obstacle $\chi = \min\{\frac{1}{2}(|x_1| - \frac{1}{2}), 0\}$ at the bottom, Dirichlet at the top and on $\{1\} \times (0, 1)$, Neumann elsewhere. No exact solution.

FHS 4.4 (singularly perturbed) is not reproduced: its manufactured solution, as printed, does not satisfy the Signorini conditions on $x = 1$ (there $u > 0$ and $\partial_n u \neq 0$).

6.1 Convergence and superconvergence

Table 2 (E1). The rates are optimal for σ_h and u_h, and ν_h gains two orders for every k, also for $k = 0$ because $f \in \mathrm{P}_0$ (Section 7). The transition is smooth here ($u_s \sim r^{11/2}$).

k	$\|\sigma - \sigma_h\|$	$\|u - u_h\|$	$\|u - \nu_h\|$	finest h
0	1.99	1.00	2.99	1/64
1	2.99	2.00	4.00	1/32
2	4.00	3.00	5.01	1/32
3	5.23	4.00	6.02	1/16

Table 2: E1, uniform refinement: rates with respect to h on the two finest meshes.

6.2 Singular solutions and adaptivity

Table 3, Figure 1. Uniform refinement is limited by $u \in H^{5/2-\epsilon}$ to $N^{-3/4}$ for σ_h and $N^{-5/4}$ for ν_h (E2); in E3 the uniform errors oscillate with the relative position of x_0 in its edge. Adaptive refinement driven by η recovers the optimal rates, including the superconvergence of ν_h, for every k. The contact terms are 0.03–1.1 % of η^2 on the last adaptive mesh: the indicator is dominated by $\|\sigma_h - \nabla\nu_h\|$.

	E2 ($x_0 = 0$)			E3 ($x_0 = 1/(2\pi)$)		
k	$\|\sigma - \sigma_h\|$	$\|u - \nu_h\|$	η	$\|\sigma - \sigma_h\|$	$\|u - \nu_h\|$	η
0	−1.00	−1.49	−1.00	−1.01	−1.51	−1.01
1	−1.51	−2.01	−1.51	−1.51	−2.02	−1.51
2	−2.03	−2.57	−2.03	−2.04	−2.57	−2.03
3	−2.57	−3.07	−2.56	−2.65	−3.17	−2.62
uniform	−0.75	−1.25	−0.75	oscillating		

Table 3: E2 and E3, adaptive refinement driven by η: slopes s in $e \sim N^s$. Optimal: $-(k+2)/2$ for σ and η, $-(k+3)/2$ for ν_h.

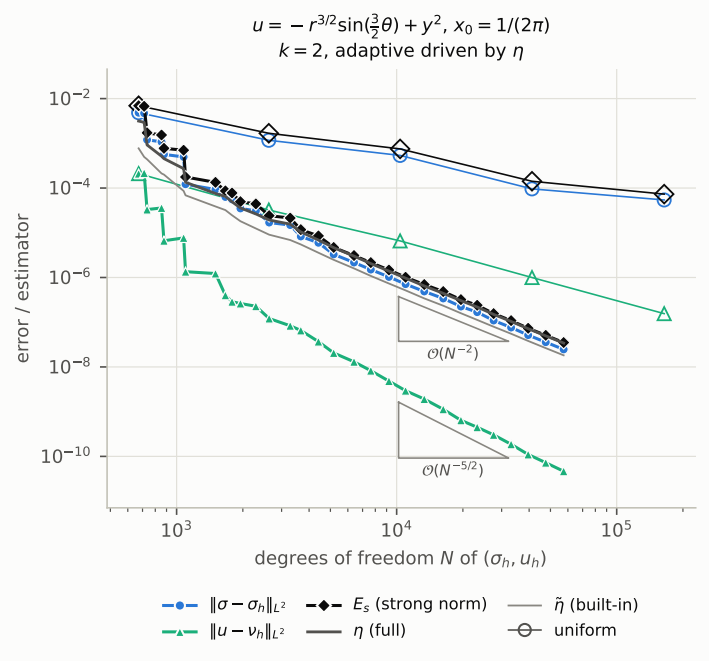


Figure 1: E3, $k = 2$: errors, η and $\tilde{\eta}$ against N; η-driven adaptive (filled) and uniform (hollow) refinement.

6.3 Choice of the contact term

On the same solutions of E2–E4 (all with $f \in \mathrm{P}_0$, so nothing but the contact can be missed), the contact term of Definition 4.1 (B) was replaced by: (A) nothing; (Bf) the same with an edge-wise classification (an edge is in contact iff all its coefficients are) instead of β_E; (Bs) the same with contact where $\sigma_h \cdot n > 0$; (C) B plus $|\langle \sigma_h \cdot n, \nu_h \rangle_E|$; (D) B plus $h_E \|\sigma_h \cdot n\|^2_{E \cap \{\nu_h > 0\}}$.

Table 4, Figures 2 and 3. With the transition at a node all variants except C and D coincide: the active set resolves the contact and the term barely matters. With the transition inside an edge and uniform refinement, B improves on A (for $k = 3$, per level, from 0.23–0.47 to 0.41–0.50), Bf improves less, Bs is noisy because $\sigma_h \cdot n$ is close to zero near the transition, C grows with k as expected from its order, and D is useless. The contact term of B follows the trace error on the

exact contact zone within a factor 0.97–2.2. E4 gives the same numbers as E3, which checks the treatment of χ.

variant	E2, adaptive	E3, adaptive	E3, uniform
A none	0.67–1.03	0.65–1.02	0.26–0.80
B Definition 4.1	0.67–1.03	0.67–1.02	0.47–0.83
Bf edge-wise classification	$\approx$ B	$\approx$ B	0.34–0.83
Bs classification by $\sigma_h \cdot n > 0$	$\approx$ B	0.67–4.45	0.63–2.58
C B + complementarity	0.68–34	0.68–29 (E4: 51)	0.49–1.14
D B + flux where $\nu_h > 0$	$3 \cdot 10^2$–$2 \cdot 10^7$	$2 \cdot 10^2$–10^7	$7 \cdot 10^1$–$5 \cdot 10^2$

Table 4: η/E_s for the contact-term variants, $k = 0, \ldots, 3$ together.

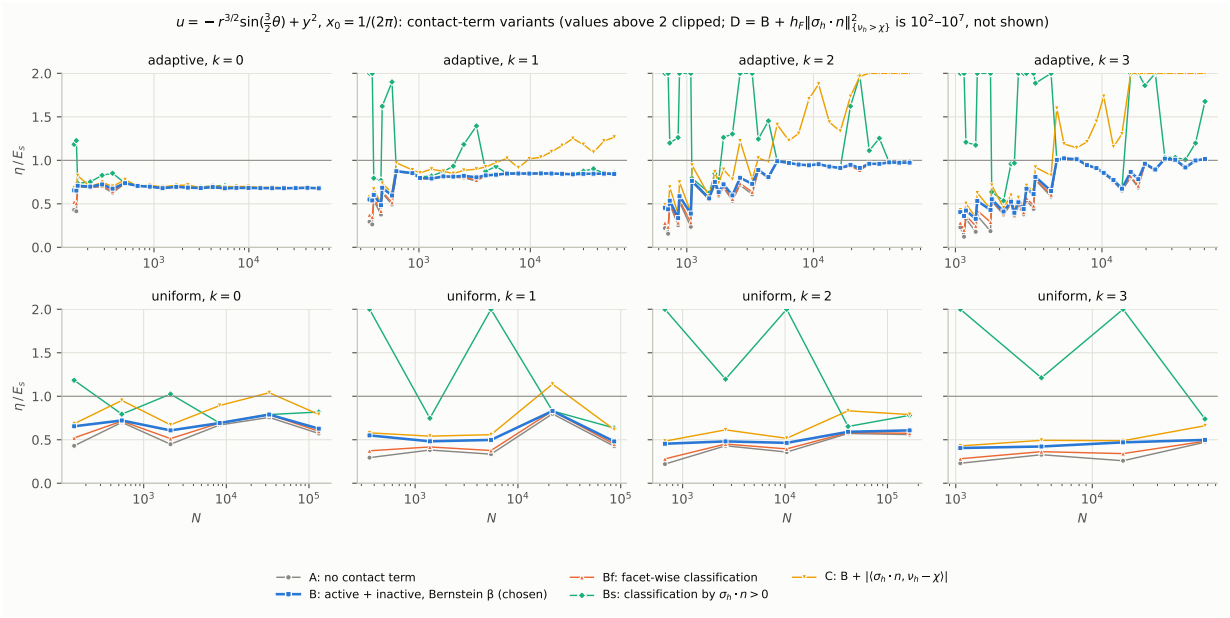


Figure 2: E3: η/E_s for the variants A, B, Bf, Bs and C; adaptive (top) and uniform (bottom) refinement; values above 2 are clipped.

6.4 Effectivity

Table 5, Figure 4. Compared with $\tilde{\eta}$, η roughly doubles the effectivity, approaches 1 for $k \geq 2$ under adaptive refinement, and converges at the rate of E_s.

example	η/E_s adaptive	η/E_s uniform	$\tilde{\eta}/E$ adaptive
E2 (x_0 a node)	0.67–1.03	0.64–0.73	0.31–0.75
E5, same u without contact	0.67–1.03	0.64–0.73	–
E3 (x_0 inside an edge)	0.67–1.02	0.47–0.83	0.31–0.73
E5, same u without contact	0.68–1.03	0.63–0.76	–
E6, FHS 4.2 ($k \leq 2$)	0.24–0.74	0.63–0.79	0.09–0.46
E8, BGK 7.2	0.20–0.32	0.35–0.73	0.08–0.17

Table 5: Effectivities, $k = 0, \ldots, 3$ together.

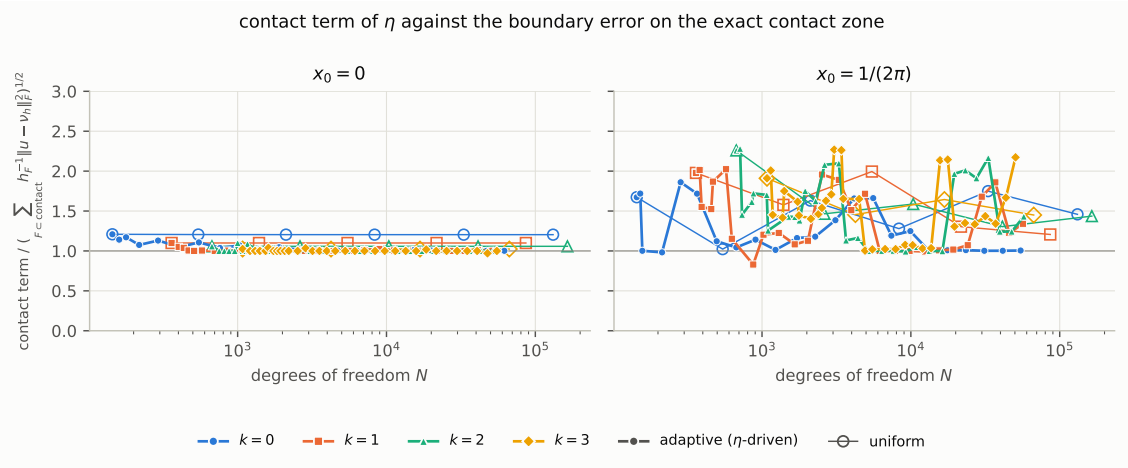


Figure 3: E2 (left) and E3 (right): contact term of η divided by the trace error on the exact contact zone, $(\sum_E h_E^{-1}\|u-\nu_h\|^2_{E\cap\Gamma_C^0})^{1/2}$.

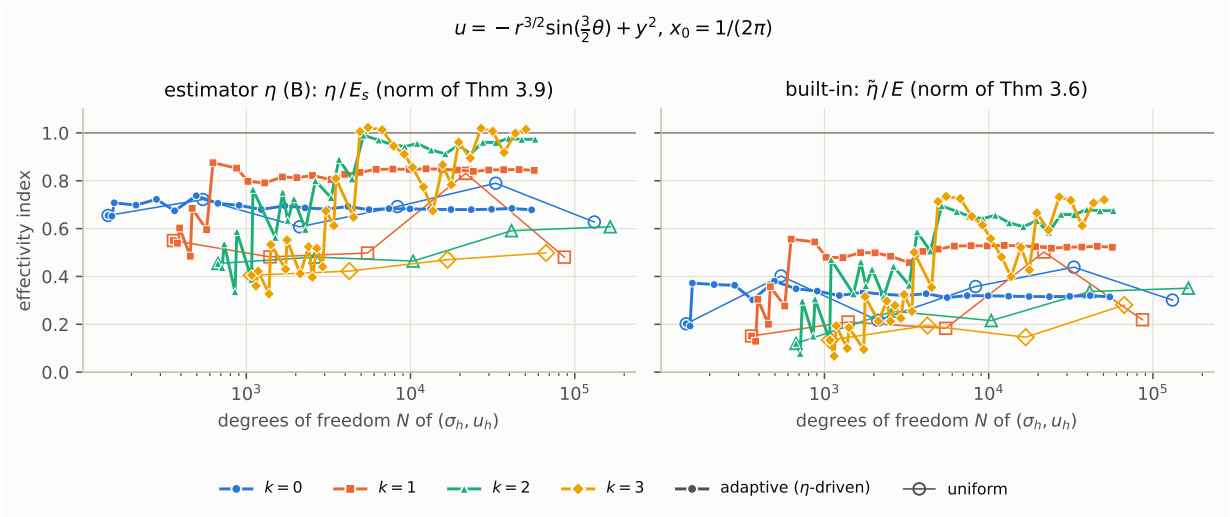


Figure 4: E3: η/E_s (left) and $\tilde{\eta}/E$ (right); η-driven adaptive (filled) and uniform (hollow) refinement.

6.5 Contact or singularity: the control without contact

Table 6, Figure 5. With the transition at a node, Signorini and the problem without contact give the same effectivities. With the transition inside an edge, the effectivity drops and oscillates under uniform refinement also without contact: part of the loss comes from the singularity lying inside an element. For $k \geq 1$ Signorini is below the control on 13 of the 14 levels, by 3–31 %; that part is a contact effect. For $k = 0$ neither is consistently lower. Under adaptive refinement the two agree within 0.03 for $k \leq 2$; for $k = 3$ the minima are 0.67 (Signorini) and 0.77 (control), and the means 0.92 and 0.97.

k	E3, Signorini						E5, same u without contact					
0	0.65	0.72	0.61	0.69	0.79	0.63	0.65	0.74	0.70	0.68	0.71	0.76
1	0.55	0.48	0.50	0.83	0.48		0.68	0.61	0.65	0.74	0.70	
2	0.45	0.48	0.46	0.59	0.61		0.50	0.57	0.50	0.75	0.63	
3	0.41	0.42	0.47	0.50			0.49	0.55	0.65	0.67		

Table 6: η/E_s per level under uniform refinement ($h = 1/4, 1/8, \dots$), transition inside an edge. With the transition at a node (E2 against E5) the two columns agree within 0.01: 0.64, 0.71, 0.71, 0.73 for $k = 0, \dots, 3$.

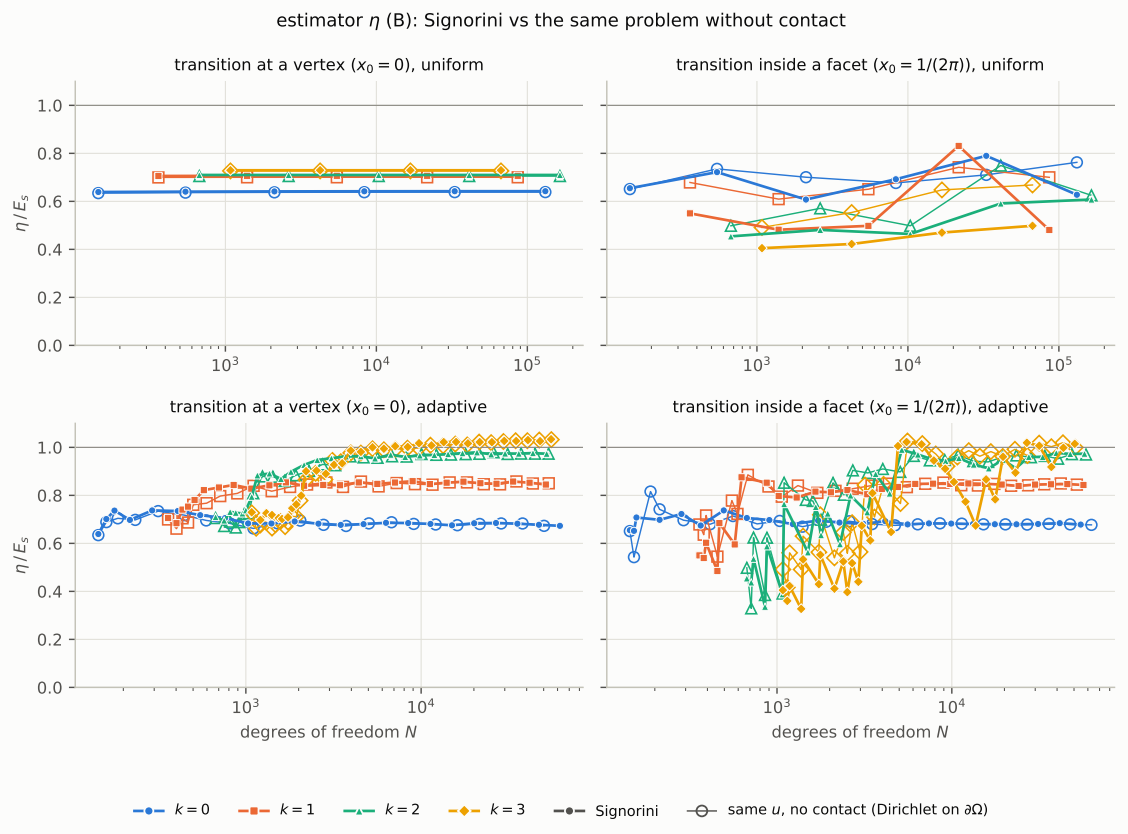


Figure 5: η/E_s for Signorini (filled) and for the same u without contact (hollow): x_0 a node (left) and inside an edge (right), uniform (top) and adaptive (bottom) refinement.

6.6 Data not in P_k

Table 7, Figures 6 and 7. In E8 the effectivity is low on the adaptive meshes and for $k = 0$, but not under uniform refinement for $k \geq 2$ (0.71–0.73). On the adaptive meshes $\delta \approx 0.93$–0.99, so the constant $1/(1-\delta)$ of [4, Thm. 3.9] is large. For $k = 0$ the oscillation has the size of the error

(osc/$E_s \approx 1.1$); for $k \geq 1$ on the adaptive meshes it is 2–4 times the error, so it does not by itself account for what is missing. E6 shows the same pattern for $k = 0$ (0.24–0.33 adaptive against 0.63 uniform, $\delta \approx 0.97$–0.99) and a much weaker one for $k = 1, 2$ (0.60–0.74 against 0.70–0.79).

	adaptive			uniform		
k	η/E_s	osc$/E_s$	δ	η/E_s	osc$/E_s$	δ
0	0.20–0.21	1.13–1.15	0.99	0.35–0.39	0.99–1.05	0.97
1	0.24–0.29	2.05–2.13	0.97–0.98	0.63–0.69	0.45–1.11	0.84–0.87
2	0.26–0.31	2.90–3.05	0.95–0.97	0.71	0.05–0.24	0.84
3	0.23–0.32	3.71–3.99	0.93–0.96	0.73	0.06–0.42	0.85

Table 7: E8 (BGK 7.2): effectivity, relative oscillation osc $:= (\sum_K (h_K/\pi)^2 \|f - Q_h f\|_K^2)^{1/2}$ and saturation constant δ. The values of δ come from the runs driven by $\tilde{\eta}$.

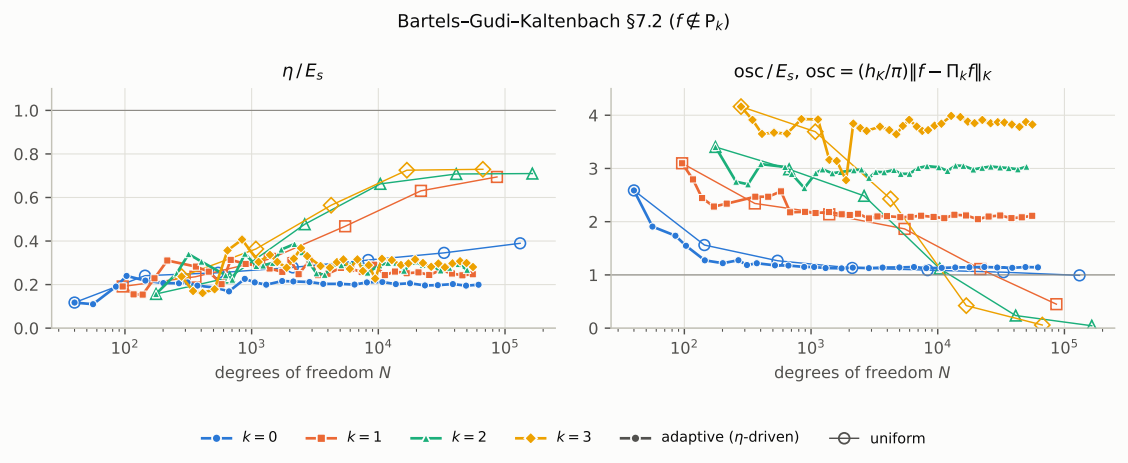


Figure 6: E8: η/E_s (left) and osc$/E_s$ (right); η-driven adaptive (filled) and uniform (hollow) refinement.

6.7 Examples without exact solution

Table 8, Figure 8. In E7 the uniform slope -0.34 matches the reduced rate reported in [2] for their lowest-order DPG method ($(\#\mathcal{T})^{-\alpha/2}$, $\alpha \approx 0.7$). Adaptive refinement gives the optimal -0.98 for $k = 0$ but stays below the optimal rate for $k \geq 1$; the meshes refine at the re-entrant corner and along the circle $|x| = 0.8$ where f jumps. In E9 adaptive refinement gives the optimal rate for every k, refining at the junction $(1, 0)$ of Γ_D and Γ_N and at isolated points of the contact boundary.

	E7, FHS 4.3		E9, BGK 7.3.1	
k	uniform	adaptive	uniform	adaptive
0	-0.34	-0.98	-0.25	-0.99
1	-0.34	-1.31	-0.25	-1.49
2	-0.34	-1.46	-0.25	-2.03
3	-0.34	-1.47	-0.26	-2.69

Table 8: E7 and E9: slope s of $\eta \sim N^s$. Optimal: $-(k+2)/2$.

Solver. The active set terminated in every run (at most 44 iterations from the fully separated state); no cycling was observed.

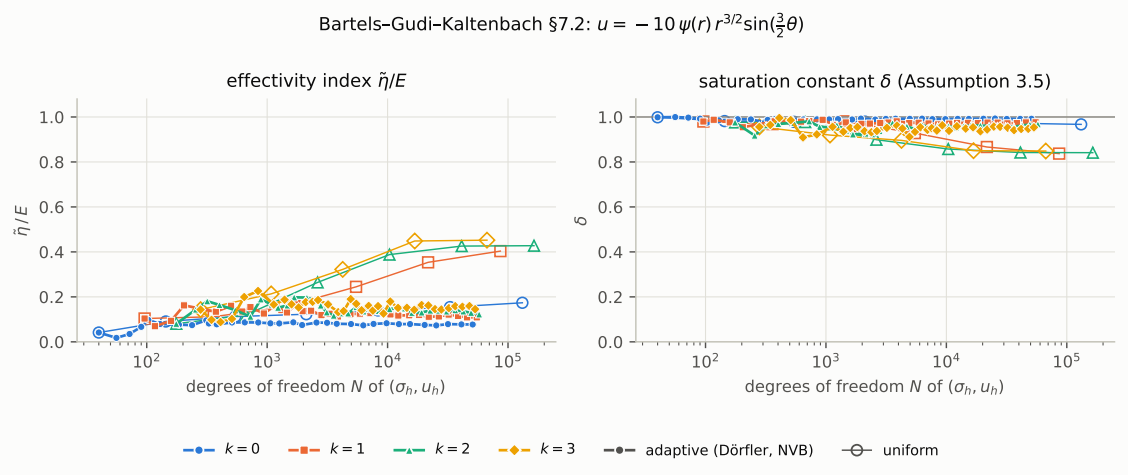


Figure 7: E8, runs driven by $\tilde{\eta}$: $\tilde{\eta}/E$ (left) and the saturation constant δ (right).

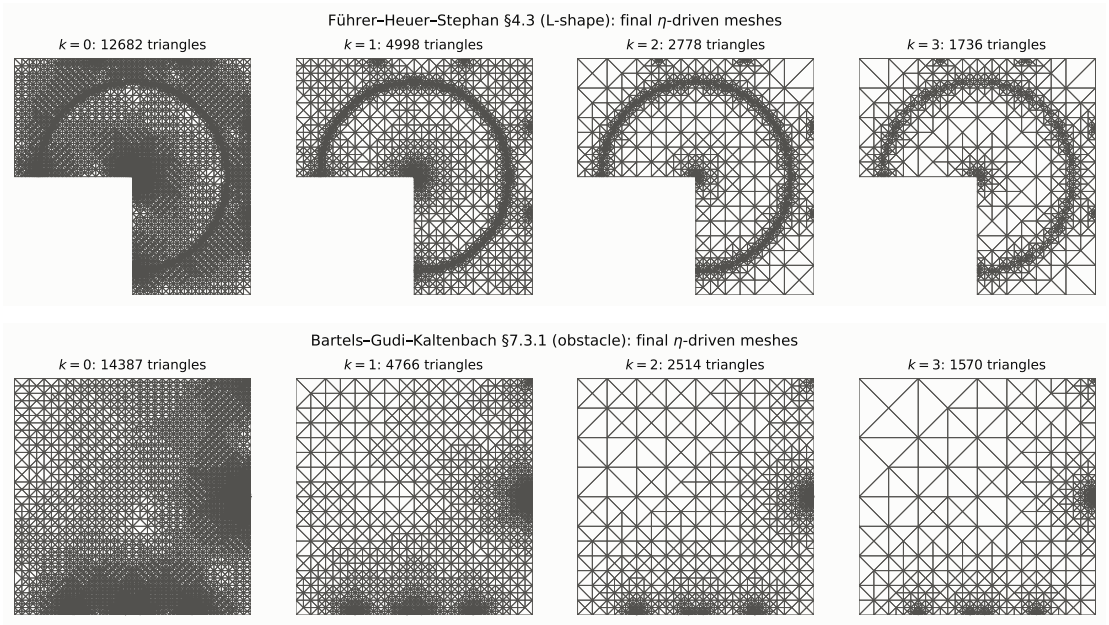


Figure 8: Final η-driven adaptive meshes for E7 (top) and E9 (bottom), $k = 0, \ldots, 3$.

7 Superconvergence at the lowest order

MRV gain *one* order for $p = k+1 = 1$ instead of two, unless $f \in \mathrm{P}_0$ [4, (2.9), (2.15)]. The reason is the bound [4, (2.11)],

$$\|u - \nu_h\|_K \le \tfrac{h_K}{\pi}\big(\|q - q_h\|_K + \|\nabla(u - Q_K^{p+1}u)\|_K\big) + \|u_h - Q_K^{p-1}u\|_K + \|u - Q_K^{p+1}u\|_K,$$

in which every term is $O(h^{p+2})$ except the supercloseness $\|u_h - Q_h^{p-1}u\|$, obtained by duality [5] on convex domains:

$$\|Q_h^{p-1}u - u_h\| \;\lesssim\; h\,\|q - q_h\| + h^{\min(p,2)}\,\|\operatorname{div}(q - q_h)\|, \qquad \operatorname{div}(q - q_h) = f - Q_h^{p-1}f.$$

For $p \ge 2$ the second term is $h^2 O(h^p)$: two orders are gained. For $p = 1$ and $f \notin \mathrm{P}_0$ it is $hO(h) = O(h^2)$: one order. For $p = 1$ and $f \in \mathrm{P}_0$ it vanishes and $h\|q - q_h\| = O(h^3)$ remains: two orders. E1 has $f = -2$, so $k = 0$ is the third case. Replacing y^2 by $y^2\cos(\pi x)$ (same conditions on Γ_C, $f = (\pi^2 y^2 - 2)\cos(\pi x) \notin \mathrm{P}_0$):

f	k	rates $\|u - u_h\|$	rates $\|u - \nu_h\|$
$-2 \in \mathrm{P}_0$	0	0.99, 1.00, 1.00	2.95, 2.97, 2.99
$(\pi^2 y^2 - 2)\cos(\pi x) \notin \mathrm{P}_0$	0	0.98, 1.00, 1.00	2.39, 2.21, 2.11
$(\pi^2 y^2 - 2)\cos(\pi x) \notin \mathrm{P}_0$	1	1.94, 1.98, 2.00	4.01, 4.00, 4.00

The theory is for the linear problem; with contact this is a numerical observation in a favourable case (smooth solution, transition at a node).

8 Open items

1. Reliability of η for the variational inequality. Even in the linear case MRV's proof uses saturation, and E8 shows δ close to 1 on adaptive meshes.
2. *Transition inside an edge.* Under uniform refinement and $k \ge 1$, Signorini loses up to 30 % of effectivity with respect to the same solution without contact (Section 6.5). None of the tested contact terms captures it, and where this error is located has not been determined.
3. *Data not in* P_k. The low effectivity on adaptive meshes (E8, E6) is compatible with a large saturation constant; the oscillation of f explains it for $k = 0$ but overshoots for $k \ge 1$. The cause is not established.
4. For $k \ge 1$ the Bernstein cone is only sufficient for $\sigma_h \cdot n \ge 0$; its effect on accuracy near transitions has not been quantified.
5. E7 with $k \ge 1$: adaptive rates below optimal, presumably limited by the jump of f along $|x| = 0.8$.
6. Variable diffusion coefficient, robustness in the reaction coefficient (FHS 4.4), 3D.

Code. The Julia package `SignoriniPostprocessing.jl` (Gridap) accompanies this note; its studies (convergence, adaptivity, references, estimators, control without contact) reproduce all tables and figures.

References

[1] S. Bartels, T. Gudi, A. Kaltenbach. A priori and a posteriori error identities for the scalar Signorini problem. arXiv:2407.10912 (2024).

[2] T. Führer, N. Heuer, E. P. Stephan. On the DPG method for Signorini problems. arXiv:1609.00765 (2016).

[3] M. Hintermüller, K. Ito, K. Kunisch. The primal-dual active set strategy as a semismooth Newton method. *SIAM J. Optim.* 13 (2003) 865–888.

[4] I. Muga, S. Rojas, P. Vega. An adaptive superconvergent mixed finite element method based on local residual minimization. *SIAM J. Numer. Anal.* 61(5) (2023) 2084–2105, doi:10.1137/22M1526307. Equation numbers refer to this version.

[5] R. Stenberg. Postprocessing schemes for some mixed finite elements. *ESAIM: M2AN* 25 (1991) 151–167.

[6] Q. Wang, J. Hu, G. Zhou. The mixed method with two Lagrange multiplier formulations for the Signorini problem. *J. Comput. Appl. Math.* 452 (2024) 116115.